PromoFinder: finding promos in microdrama scripts

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App https://promofinder-rb4wbkeye8i8jttfu7ay67.streamlit.app/ · **Code** https://github.com/Harsh-2005-Bhati/promofinder · **Raw outputs** github .../outputs

Why this problem

An editor watches every episode to pick promo moments, and Kuku releases new shows every week. That is a repeatable, measurable bottleneck: a promo either has a hook in 3 s, a cliffhanger at the cut and makes sense alone, or it doesn't, and its timestamps can be checked exactly. The screenplays already carry a timestamp for every line, so most of the work can be code, with AI only where judgement is needed.

Assumptions (where the brief was silent)

- **Promos recruit new viewers, who start at episode 1.** So a late-series reveal ("You're the Duke?") is a spoiler, not a great promo, even if it is the most dramatic line in the show.
- Script timestamps match the final video. Hook = first 3 s, cliffhanger = last 5 s, strict 20–60 s.
- Default promo = one continuous clip; I also build two-part "cold opens" (a ≤6 s teaser line + a main clip) from the same episode. Cross-episode joins are left for later.
- Free tiers only: the free Gemini tier may use inputs to improve Google's products, so **Kuku's scripts never went to a free LLM**; the LLM was run on a synthetic pack. No script files are published; outputs quote only the single hook/ending line needed to review each clip. Volume assumed: 10 shows a week.

How it works: what is code, what is AI, where the human sits

- **Code owns every timestamp.** A parser reads all 50 scripts (3,727 blocks), candidate clips are cut only at line boundaries (never mid-dialogue): 16,167 windows of 20–60 s.
- **Rules score each clip** (free, ≈18 s for the whole pack): hook, cliffhanger (judged on the last *line*, so a music swell can't fake one), standalone (mid-conversation openers, a slap just before the cut), density, and **spoiler risk** weighted by how late the episode is. Top 5 per episode after de-duplication, global ranking with ≤2 per episode in the top 20.
- **Optional LLM judge** (one call per episode) sees numbered lines and may only return *block indices* and 1–5 scores, never times. Any model plugs in: Gemini, Anthropic, OpenAI, Groq, or **any OpenAI-compatible URL** (`--llm custom --base-url URL --model NAME`), with cache, retry and rule fallback.
- **A validator checks every promo** (length, bounds, no cut mid-line, a line in the first 3 s, and that the quoted hook/ending text is really inside the clip). Failures are flagged, never silently fixed.
- **Human in the loop:** Approve/Reject with a reason in the app → `eval/labels.csv` → `evaluate.py` and `feedback.py` tell me what to fix next. Output is a ranked list plus an editor pack (clip sheet with timecodes, briefs, EDL for Premiere/Resolve, ffmpeg cut scripts).
- Input can be a ZIP, a folder or single .txt episodes; 10 kinds of broken files (uppercase, missing headers, renamed characters, HH:MM:SS times, truncated, garbage) produce warnings, never a crash.

What I'd post first, and why

#	Clip	Opens on (first 3 s)	Cuts on	Why it works
1	Ep 1 00:46.8–01:10.6 (24 s)	"You knew this before I walked through those doors!"	"And you knew?"	Altar betrayal: the show's premise in 24 s, no context needed
2	Ep 27 00:44.5–01:32.7 (48 s)	"What is your husband's name?" + doors pounding	"…knew better than every physician in this house?"	A secret about to come out, interrupted
3	Ep 32 01:59.6–02:34.4 (35 s)	"Silence! Silence!"	Juliana gasps… the glowing tip of the iron thrusts forward	Highest stakes; it's the episode's own ending
4	Ep 3 00:25.9–01:17.1 (51 s)	"You have humiliated this family enough for one day."	"And who are you to give orders in my family's chapel?!"	A mysterious stranger steps in
5	Ep 21 00:11.2–00:35.8 (25 s)	"My little brother… cannot breathe!"	"…he will die! You are his only hope"	A ticking clock with a child's life

All five were approved in review, come from five different episodes, and all have near-zero spoiler risk. Earlier episodes are favoured on purpose: a new viewer who clicks starts at episode 1.

How I know it's good (real runs on the full pack)

Check	v1 (blueprint rules)	v2 (after review)
Top-20 precision (approve/reject with reasons)	11 / 20 = 55 %	**16 / 20 = 80 %**
Key moment (15 episodes, picked blind from the logline) in the episode's top 3 / top 5	13 / 15 · 15 / 15	11 / 15 · **15 / 15**
Validator pass: single promos · cold opens	196 / 196 · 40 / 40	**201 / 201 · 41 / 41**
LLM judge (Gemini 3.8 Flash, synthetic pack)	–	6/8 episodes judged (2 timeouts → rule fallback); 13/14 range changes accepted; agrees with rules on 5 of top 10

Round 1 rejects fell into four patterns: spoilers, fake cliffhangers (a calm or tender ending with a music swell), openers that only make sense mid-conversation, and a clip starting just after the slap that explains it. I turned each into a rule, re-ranked, and reviewed the 12 *new* top-20 clips blind, then froze the rules so I wasn't fitting them to my own labels. **Trade-off:** v2 is more precise but found fewer key moments in the top 3 (13 → 11 / 15). Partly that is correct (v2 ends on the question and leaves the answer out), and partly a real regression (below).

Where it picks badly (verbatim)

- **Opens without context:** ep 32 clip opens on "Burn it." (burn *what*? the book is introduced in the previous scene); ep 29 opens on "You still call this a legacy?"

- **Loud ≠ tense:** ep 29 ends on Juliana's lament in the cell ("I saved her! I healed her!"), so the rules read shouting as a cliffhanger.
- **Spoiler the word list misses:** ep 43 title-fraud exposition ("declared dead", "title") reached #20.
- **Regression:** the ep 39 finale ends on an action line ("…eyes burn with furious determination as the screen cuts sharply to black"); v2 only rewards "cut to black" after a tense line, so the best ending in the episode scored 0 for cliffhanger.
- **Learning from the reviews:** the cliffhanger, context and spoiler scores now clearly separate approved from rejected clips, but the hook score barely does (0.91 vs 0.82). So the next work goes into hook rules, not weights.

Scale and cost

Everything except the LLM judge is plain code: ≈18 s and ₹0 per 50-episode show, linear in episodes. The judge costs one call per episode: measured ≈2,550 input tokens; output 1,000–4,000 tokens depending on thinking. At Gemini 3.8 Flash paid prices (2027 list, ₹95/$) that is **₹1.1–3.2 per episode, ₹54–161 per show, ₹270–800 a week at 10 shows with the Batch API**. Latency, not cost, is the constraint (18.6 min for 8 episodes on the free tier), so production would batch or parallelise, and the cache makes re-runs free. A human still approves before anything is posted.

AI tools I used, and where they got it wrong

I used **Claude (Claude Code)** to build and test the code, run the audits and draft the review decisions from each clip's full script, which I then checked; and **Gemini 3.8 Flash** as the optional judge. **Where they got it wrong:** (1) Gemini rejected three clips, proposed better ranges, but quoted lines from the *old* range. The validator caught all three ("text not in clip"); the cause was my prompt not saying which range to describe, now fixed with a regression test. (2) Claude's first cost estimate ("under ₹1 per episode") ignored Gemini's thinking tokens, which are billed as output; the real figure is ₹1–3. (3) A keyword rule matched "fire" inside "firewood", which put a weak clip in the top 20 until review exposed it.

Limits and what I'd do next

Two review rounds of 20 by me (with AI drafting), on one period-romance series: a blind review by the promo team is the real test. Next: (1) sit with the promo team and A/B the top picks against hand-picked promos on 3-second hold and click-through; (2) fix hooks and the ep 39 regression, then re-review; (3) run the LLM judge on the real pack with a no-training key and keep it only where it beats the rules; (4) cross-episode promos; (5) speech-to-text + shot detection so shows without timed scripts produce the same input.